

Abstract No.

Characterizing TDO2 as a heterogeneous mesenchymal-associated metabolic vulnerability within the tryptophan–kynurenine axis in glioblastoma

Miriam Meloni^{1,2,*}, Ilaria Pagano², Thomas Daubon³, and Rossella Galli²

¹Vita-Salute San Raffaele University

²San Raffaele Scientific Institute

³University of Bordeaux

*Presenting author

Background – Despite therapeutic advances, glioblastoma (GBM) remains incurable, partly due to its extensive cellular and molecular heterogeneity. The mesenchymal (MES) phenotype, in particular, is associated with aggressive features, inflammatory signaling, and an immune-enriched microenvironment. Although the tryptophan-kynurenine pathway has been implicated in GBM, its association with subgroup-specific GBM stem cell states, its inter- and intra-tumor heterogeneity, and its functional relevance in patient-derived models remain poorly defined.

Methods – Patient-derived GBM stem cells (GSCs) were cultured in vitro and used for xenograft generation. Kynurenine pathway components were evaluated by immunohistochemistry and western blot, while functional perturbation was performed using shRNA-mediated silencing. The effect of silencing was assessed in vitro by functional assays and in vivo using intracranial NSG xenografts that were longitudinally monitored by MRI. Bioinformatic cross-cohort analyses were performed on public bulk RNA-seq, sc/snRNA-seq, and spatial transcriptomic GBM datasets.

Results – Untargeted metabolomics of proneural and MES GSC-derived xenografts identified kynurenine accumulation as a prominent feature of the MES ones, with TDO2 showing the clearest MES-associated expression among kynurenine-producing enzymes. Across independent GBM cohorts, TDO2 progressively tracked MES identity although displaying marked inter-tumor and intra-tumor heterogeneity. Single-cell and spatial analyses localized TDO2 to specific MES-like malignant states and vascular-associated niches characterized by inflammatory programs. Functionally, TDO2 silencing in MES GSCs significantly impaired tumor growth.

Conclusions – Our findings refine the role of TDO2 in GBM, highlighting it as a candidate MES-associated metabolic vulnerability with functional relevance in patient-derived orthotopic models. Its marked inter-tumor, intra-tumor, and spatial heterogeneity suggests that TDO2 is associated with specific tumor states and niches, rather than representing a universal GBM feature. This heterogeneity may therefore inform patient stratification for future TDO2-targeted approaches.



Supported by SciSerTec

Web: <http://www.scisertec.de>